Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/29/2020
PI Name:	Sanford, Larry Ph.D.		
Project Title:	Impact of Inflight Stress and Sleep Disturbances on Brain Function, Neural Communication, and Inflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	09/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	July 2020 report: None.		
COI Name (Institution):	Britten, Richard Ph.D. (Eastern Virginia Medical School) Chuang, Shao-Hui D.Eng. (Norfolk State University) Ciavarra, Richard Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
Grant/Contract No.:	80NSSC19K1582		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts may experience several physical and psychological challenges including prolonged exposure to microgravity, potential social isolation (SI), and predicted exposure to ~13 cGy/Yr Space Radiation (SR). Each of these stressors can have deleterious effects on physical and mental health as well as negatively impact mission performance. Each also can disturb sleep, which can further exacerbate the effects of stress. Determining how astronauts respond to the single and compound effects of mission related stressors is crucial for understanding and mitigating factors that could impede mission success. However, examining their effects will be complex because they have different time courses and durations. SR exposure will be episodic whereas microgravity will be constant and SI and sleep disturbances can be temporary or prolonged. Their effects may also be additive or synergistic, and may vary with sex and with individual differences in stress resilience and vulnerability. Therefore, our goal in this project is to determine the single and synergistic effects of inflight stress and SR on mission related biological and functional challenges that astronauts may face. We will determine how order of experience impacts the effects of inflight stressors and determine sex and individual differences in stress resiliency using an established Wistar rat model. We will use established ground-based models of microgravity (hindlimb unloading), SI, and SR. We will use telemetry to record sleep and to assess temperature increases as an index of the stress response. We will use plasma levels of brain derived neurotropic factor to predict individual differences in stress responses. We will assess fear learning and extinction to determine how chronic inflight stressors impact ability to respond to an additional acute stressor. We will analyze food entrained, anticipatory activity to assess potential impairments in temporal perception. We will determine the effects of inflight stress and SR on sensorimotor function and recognition memory using behavioral tests. We will use local field potential recordings in the amygdala, medial prefrontal cortex, and hippocampus to determine whether stressors and SR alter the coordinated neural activity underlying fear learning and extinction, and we will determine whether there are structural changes within connections between regions. We will use chemokine/cytokine arrays and Ingenuity Pathway Analysis to determine the effect of stressors and SR on the immune system. Collectively these studies will assess functional, mechanistic, and immune consequences of several stressors inherent in space exploration. The data they provide will aid in risk assessment, and may help to develop countermeasures based on individual differences in ability to withstand stress and/or targeting immune pathways that may confer protection against the deleterious effects of multiple stressors that astronauts will face.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will increase understanding of how individual differences in resilience and vulnerability factor in determining lasting outcomes of stressful experiences. They will also determine the role that stress-induced alterations in sleep play in mediating the outcomes of stress. The results will have importance for understanding how individuals cope with stress on Earth and on missions in space.
Task Progress:	Our goal in this project is to determine the impact and potential synergistic effects that multiple inflight stressors and exposure to SR can have on biological systems and mission related performance. Outcomes that we will examine include sensorimotor function and balance (adhesive removal task and balance beam performance), recognition memory (novel object recognition task), time perception (anticipatory activity), fear memory and fear memory related neural communication. We will also examine the relationship of altered performance to the effects of inflight stressors on sleep, and how responses to stressors vary in resilient animals that maintain sleep after experiencing stress and vulnerable animals that have reduced sleep after experiencing stress. This is the first year of this project. During the Definition Phase, methodologies were standardized and a cross-species validation study was designed to compare the effects of GCRsim (galactic cosmic radiation) on sleep and behavior in mice and rats. Pilot studies were conducted to optimize anticipatory activity and adhesive removal tasks in rats. We conducted one study (n=6) to determine EEG changes associated with anticipatory activity. We implemented the cross-correlation analysis need to assess coordinated activity between local field potentials recorded in the amygdala, medial prefrontal cortex, and hippocampus in our studies on neural communication. These efforts, thus far, have found that sleep resilient and vulnerable rats show different patterns of cross-correlated activity in the amygdala and hippocampus in the aftermath of footshock stress and after fear memory recall. Specifically, the cross-correlated activity between local field potentials in the amygdala and hippocampus is reduced during rapid eye movement sleep and non-rapid eye movement sleep in resilient rats, but not in vulnerable rats. Studies will be conducted to determine whether space flight stressors produce adverse effects on neural communication that is associated with alterations in behavioral performance and emotion. As we were scheduled to enter the Implementation Phase of the project, work was halted due to the COVID-19 pandemic shutdown. One cohort of rats (n=23) has been implanted for recording sleep via telemetry and will begin behavioral testing in the last two months of the current grant period. These tests will include anticipatory activity, novel object recognition, adhesive removal, balance beam and fear conditioning, and extinction. Additionally, we will determine whether the effects of a mild stressor on sleep are predictive of those of intense stress.
Bibliography Type:	Description: (Last Updated: 03/16/2026)
Abstracts for Journals and Proceedings	Sanford LD, Wellman LL, Machida M, Sweeten BLW, Britten RA. "Modeling differential effects of space radiation on sleep using stress vulnerable and resilient rats." Presented at the 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020